Representational identity across role changes

A governance layer beneath the claim-state ledger (Doc. 303)

Johann Pascher

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Contents

.1 Roles, not strengths 2
.2 The failure mode: role drift 3
.3 Case study I (external): one number in four roles 3
.4 Case study II (internal): the information unit 3
.5 Definition identifies, governance preserves 4
.6 The identity-preservation record 4
.7 Relation to the ledger 5
.8 Status 5

Doc. 303 Representational identity across role changes — a governance layer beneath the claim-state ledger

The claim-state ledger asks whether a transition is *earned*: whether an assertion may be booked as proven, restricted, or blocked. That test is preceded by an earlier question it does not ask — whether the object undergoing the transition has kept its *representational identity*. A role change is not a strength change. The same symbol, the same number, the same structure can successively occupy different representational roles — mathematical object, computational output, candidate physical quantity, observable, ontological posit — and it is between those roles that the real, often unnoticed, shift takes place. This document names that layer, gives two case studies (one separates the definition (which identifies) from the governance (which preserves), and proposes an identity-preservation record as a layer prior to the ledger.

.1 Roles, not strengths

A representational object can occupy at least the following roles within a framework:

- *mathematical object* — a quantity within a formal structure, with no physical attribution;
- *computational construct / output* — the deterministic result of a procedure;
- *candidate physical quantity* — an object to which a physical meaning is *proposed* but not yet secured;
- *observable* — a quantity coupled to measurement;

- *ontological posit* — a claim about what *is*, beyond derivation and falsification.

These roles are not stronger or weaker versions of the same claim; they are different *types*. The claim-state ledger sorts assertions by evidential strength (proven / restricted / blocked); the role question sorts objects by their representational type. Both axes are needed, and they are orthogonal.

.2 The failure mode: role drift

The typical error is not the wrong evidential strength but *silent role drift*: the same numerical or mathematical object changes its type without the change being declared as such. A number produced as a lattice output is labelled a "bare" physical mass, then carried as a candidate observable, and finally treated as an observed value after a correction still to come. These are four representational roles of the same numerical object. None of them is inadmissible in itself; what is inadmissible is only to present the role change as a mere strengthening of the same claim. This is precisely where disagreement enters quietly — often long before the evidential question is even asked.

.3 Case study I (external): one number in four roles

A numerical object from a discrete lattice model passed, in an IPI discussion, through four roles: first a *lattice output* (a deterministic computational result), then a *bare topological mass* (a physical attribution), then a *candidate* for an observable ratio, and finally an *observable after screening* (under a reduction mechanism not yet constructed). The progress of the discussion consisted not in refuting any one of these roles but in *separating* them: once the roles were laid apart, each could be examined on its own constitutional footing — the lattice output as proven, the physical attribution as a definition, the observable as a still-open transition. Separating the roles was the actual methodological gain, independent of the fate of the model.

.4 Case study II (internal): the information unit

The same role drift occurred, in the preparation of the present document series, within FFGFT itself. The question of the "minimal information unit" was answered successively with four different objects: first a *spectral eigenmode* (an object of the mass operator), then an *area* (the holographic choice of one bit per area element), then a *counting quantity* of order $1/\xi$, and finally the *topological winding quantum* $\Delta w = 1$. These four are not stronger and weaker versions of the same unit; they are four different representational objects that successively bore the same name. The role change went unnoticed until the identity was fixed by a *prior, explicit definition*: Doc. 257 fixes the information unit, by definition, as the winding quantum $\Delta w = 1$ ($\pi_1(T^4) \cong \mathbb{Z}^4$, scale-universal). Only that identity anchor made the remaining roles legible as what they are: the area as a lossy projection, the $\ln 2$ as a thermodynamic shadow, the energy $\hbar c/L$ as a scale-specific projection — all *derived or projected* appearances of the one fixed unit, not competing definitions.

.5 Definition identifies, governance preserves

The lesson from both cases seems at first to be that an *identity anchor* preserves identity. But that conflates two questions that belong apart — and the conflation is exactly the role drift this document describes, now applied to the document itself. There are two constitutional questions:

- the *declarative* question — *what is this object?* — is answered by a *definition*. The definition *identifies*: it sets the constitutional reference point against which any further claim can be checked at all. In FFGFT, $\Delta w = 1$ (Doc. 257) does this: it fixes *what* the information unit is.
- the *governance* question — *under what operations does it remain the same object?* — is different. A definition sets the initial reference; it does not *preserve* identity. Identity is preserved only by the governance imposed on subsequent transitions.

In short: *the definition identifies, governance preserves*. The earlier notion of an "identity anchor" carried both roles under one name — definition *and* preservation mechanism — and is therefore itself an instance of the drift described here. It is cleaner to separate them: the definition is the declared reference point; governance is a distinct layer that records what happens to that reference at each transition.

.6 The identity-preservation record

A merely binary check — same type before and after the transition → the ledger; different type → substitution — is too coarse. A transition can treat an object's declared identity in more than two ways, and the governance layer keeps an explicit record of which: for each transition it records whether it

- *preserves* — the same object in the same role;
- *projects* — a scale- or coordinate-dependent appearance of the same object, lossy and not fundamental;
- *transforms* — a derivable reshaping that keeps the reference;
- *substitutes* — a different object under the same name (undeclared, this is role drift; declared, it needs its own licence);
- *bridges* — a secured mapping onto an object of another framework.

Read against FFGFT's own information unit: the winding quantum $\Delta w = 1$ is *preserved*; energy ($\hbar c/L$) and area are *projections*; the $\ln 2$ is a *transformation* (the thermodynamic shadow); the earlier wandering eigenmode → area → $1/\xi$ was *substitution* under a retained name, until the definition (Doc. 257) fixed the reference; and the verification direction FFGFT→HLV is a *bridge*. The record decides no physics; it only keeps visible, at each step, what happens to the identity — before the ledger asks whether the step is earned.

.7 Relation to the ledger

Three levels are now separated. The *definition* identifies the object (declaratively). *Governance* preserves its identity across transitions and keeps, in the preservation record, whether a step preserves, projects, transforms, substitutes, or bridges. The *claim-state ledger* scores whether a transition is *earned* (proven / restricted / blocked). Governance is prior to the ledger: it ensures that a discussion is held about the *same* object before it is held about that object's evidential strength. All three levels are largely independent of the underlying physics — they concern the bookkeeping of representation, not its content — and gain importance as independently developed frameworks come to interact. Whether this bookkeeping is kept informally or through a systematic mechanism does not change its effect: the discussion shifts away from defending positions and towards localising where assumptions reside, what has genuinely been derived, and where an object changes its identity.

.8 Status

This document is a methodological proposal, not a physical derivation. The FFGFT-specific statements — the information unit as $\Delta w = 1$, the three levels, the projection nature of energy and area — are corpus-supported (Doc. 257, 301, 300); the role taxonomy and the identity check are constitutional bookkeeping and are booked as such. The second case study is explicitly self-critical: FFGFT itself required the discipline proposed here, and the identity anchor $\Delta w = 1$ was recognised as such only in retrospect (Doc. 257). The occasion of the question — an objection to elementary cells, the constitutional framing of the transition problem, and the representational typing of the objects involved, as developed within the IPI circle (Takayuki Ueda, Stefaan Vossen, José Tomás Guevara Calderón) — is gratefully acknowledged. The distinction between declarative identity (definition) and its preservation by governance, and the five-way preservation record (preserve / project / transform / substitute / bridge), were sharpened in exchange with Stefaan Vossen; they are constitutional language made jointly visible, neither a replacement for nor an adoption of any particular governance programme.